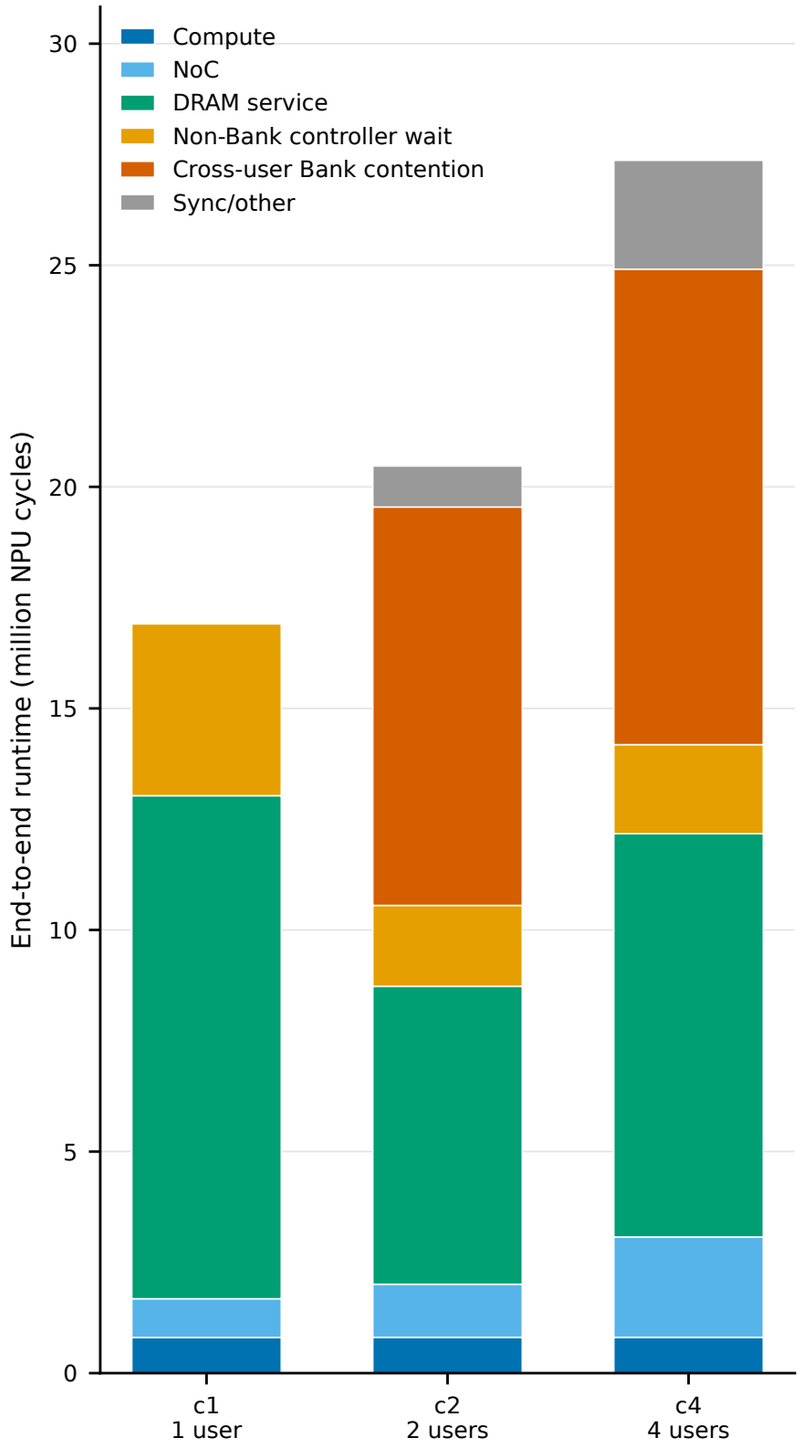
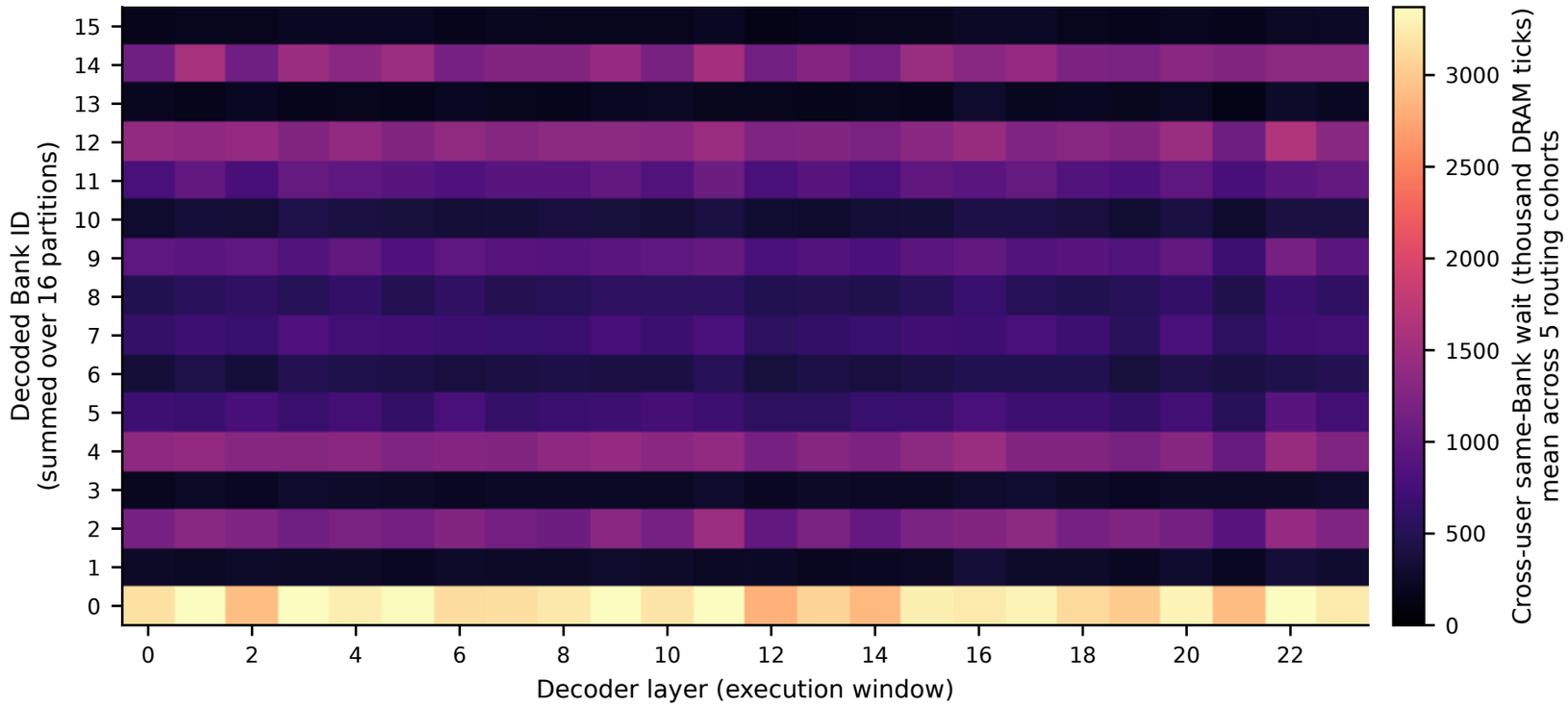
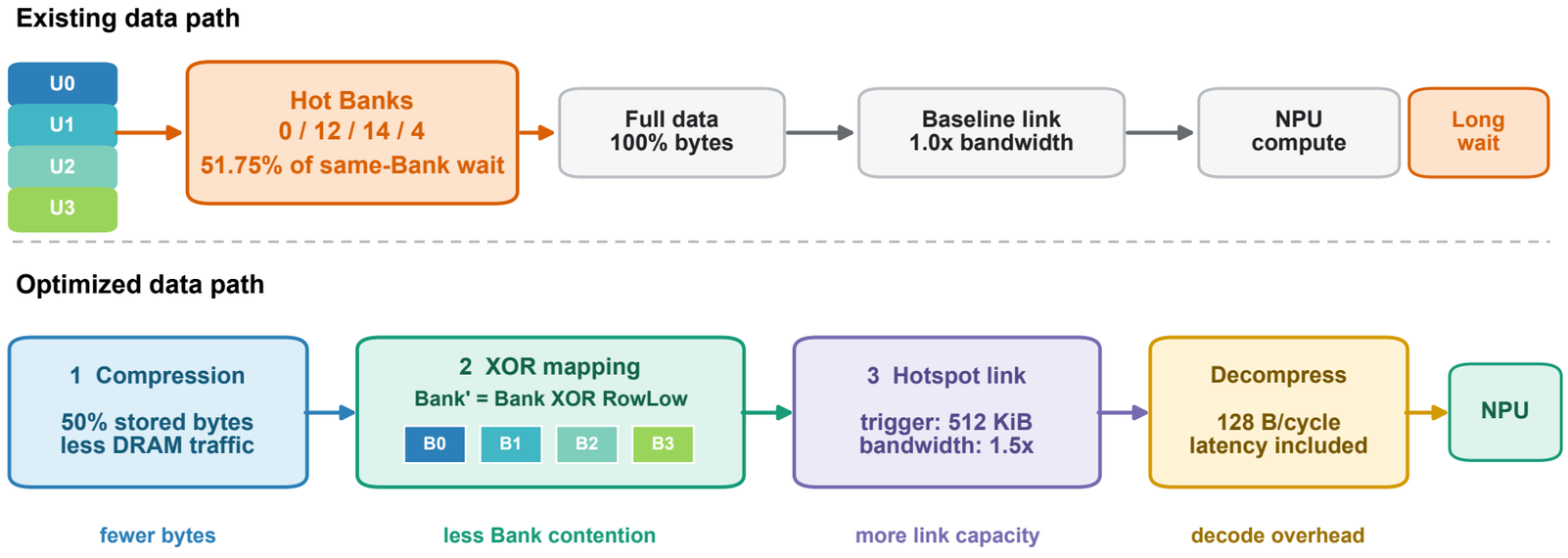
(a)


Compute
NoC
DRAM service
Non-Bank controller wait
Cross-user Bank contention
Sync/other
End-to-end runtime (million NPU cycles)
0
5
10
15
20
25
30
c1
1 user
c2
2 users
c4
4 users

(b)


Decoded Bank ID
(summed over 16 partitions)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
0
2
4
6
8
10
12
14
16
18
20
22
Decoder layer (execution window)
Cross-user same-Bank wait (thousand DRAM ticks)
mean across 5 routing cohorts
0
500
1000
1500
2000
2500
3000

(c)


Existing data path
U0
U1
U2
U3
Hot Banks
0 / 12 / 14 / 4
51.75% of same-Bank wait
Full data
100% bytes
Baseline link
1.0x bandwidth
NPU
compute
Long
wait
Optimized data path
1 Compression
50% stored bytes
less DRAM traffic
2 XOR mapping
Bank' = Bank XOR RowLow
B0
B1
B2
B3
3 Hotspot link
trigger: 512 KiB
bandwidth: 1.5x
Decompress
128 B/cycle
latency included
NPU
fewer bytes
less Bank contention
more link capacity
decode overhead